

NutrioBox Dispatch App — Full Logic Reference

Every rule the gun and worker currently run: scan validations, quantity caps, auto-post & delta, same-voucher append, Submit, sync, and completion reconciliation. Compiled from the live app + worker code, 19 Sep 2026.

A Scan-time validations (run on EVERY scan, before an item is added)

A scan is only added to the order if it passes ALL of these. Any failure buzzes, shows a message, and does NOT count the scan:

- **1. Batch in inventory** — the scanned batch must exist in the synced stock. Unknown batch → blocked.
- **2. Warehouse = ideal warehouse** — the batch's warehouse must equal the product-master's dispatch (ideal) warehouse. Stock sitting in another warehouse can't be dispatched.
- **3. Not expired** — the batch expiry must be in the future.
- **4. Quantity ≤ pending** — scanned packs for a line can't exceed what the sales order still needs.

Two further stock caps back these up:

- **Per-batch cap** — never scan more of one lot than that lot holds (race-safe: it counts an immediate per-batch tally, so a fast burst can't slip one extra past while the saved list catches up).
- **Product-total cap** — never scan more of a product than its total dispatchable stock across all lots in the ideal warehouse.

B Auto-post — the 10-minute delta cycle

While an order is open, auto-post fires every **10 minutes**. Each cycle: autosave → refresh stock from PACT → compute the delta → post it → on a confirmed success, record those packs so they're never sent again.

```
delta(line) = scanned(line) - alreadyPosted(line)    (only lines > 0 are sent; each capped at batch stock)
```

C Why the 2nd auto-post sometimes repeats the 1st (Rajinder Nagar vs Karkardooma)

alreadyPosted is updated **ONLY** when a post is confirmed by PACT. This one rule explains the two behaviours you saw — it is the same logic, reacting to two different outcomes:

	Rajinder Nagar (OMR-2267)	Karkardooma / KKB (OMR-2261)
1st auto-post result	posted= true , entered 22/22 — CONFIRMED	posted= false , entered 8/9 — NOT confirmed
So alreadyPosted...	recorded the 22 items	recorded nothing (post unconfirmed)
2nd auto-post delta	only NEW items since the 1st → no repeats	re-sent the 1st batch + new items → repeats
Why it's safe	nothing to re-send	re-send avoids losing items; worker ledger + PACT cap stop any double-dispatch

Takeaway: a repeat in the 2nd auto-post means the 1st post could not be confirmed — the app is re-sending for safety, not double-dispatching.

D Same-voucher append & the double-dispatch guard

- The 1st auto-post **CREATES** the order's voucher; the 2nd/3rd **APPEND** to that same voucher while it is still open.
- The **worker keeps a ledger** (voucher-map.json) of exactly which packs are already on each order's voucher. Even if the gun re-sends everything, the worker posts only packs not already there.
- If the voucher is already fully posted, the worker opens a fresh voucher for the remaining delta instead of appending.

E Submit (manual) vs auto-post

- **Auto-post** = background, incremental, keeps the order open, appends deltas every 10 min.
- **Submit** = the **FINAL** post: sends all remaining un-posted scanned items at once, marks the order **COMPLETED**, and stops the auto-post timer.

- If any post ends unconfirmed, auto-post **PAUSES** for safety until the operator verifies in PACT (a stock move is never repeated blindly).

F PACT's own safety cap

Independent of the gun and worker, **PACT caps every line at its own pending quantity**. It can never deliver more than what the outlet ordered — so a re-send or an over-scan can never over-dispatch.

G Sync logic (how PACT data reaches the gun)

- **New orders** (outlet requisitions + B2B) sync automatically every **3 minutes**; a manual 'Sync from PACT' pulls them on demand (~90s).
- **Inventory / stock** syncs every **15 minutes**, and on each auto-post cycle and each manual sync.
- **Dispatch jobs have priority** in the worker queue, so frequent syncs never delay a post.
- The order list on the gun re-reads every 30 seconds so new orders appear on their own.

H Completion reconciliation (PACT is the source of truth)

- When an order is opened, if PACT still shows it as pending, any stale local 'completed' lock is cleared automatically — an order can't show completed on the gun while PACT still owes stock.
- On a PACT rejection, the gun shows PACT's exact reason and source (e.g. 'Product not available in SO Dim:10') — nothing was dispatched, so it is safe to fix and re-scan.

I Known failure mode being fixed (ledger drift)

The delta logic is correct only while 'alreadyPosted' matches what PACT actually delivered. If a post is recorded done but PACT took fewer units, the ledger over-counts, later deltas shrink to ~0, and items stay pending while showing green. **The fix in progress** drives the green/pending status and the delta from PACT's live delivered quantity (ordered – pending) instead of the gun's memory.

Reference for the NutrioBox / Sukhna Foods dispatch team. Sections A–H describe the live behaviour; section I is the reconciliation improvement underway.